Muhammad Nabeel Malik

Full-Stack AI Engineer | LLM Agents & Real-Time Backends

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Professional Summary

Full-stack AI engineer with 5+ years shipping production software across industrial SaaS, fintech and e-commerce. Currently building Claire, OSPHIM's injection-molding AI agent (Python, FastAPI, Claude), live for all customers, alongside real-time multi-tenant backends in NestJS, PostgreSQL and Kafka. Strong on data modelling, honest architecture and taking AI from prototype to production.

Work Experience

Full-Stack AI Engineer, *OSPHIM* 01/2026 – Present

Full-Stack AI Engineer · Full-time 07/2026 – Present

Full-Stack Engineer · Part-time 01/2026 – 07/2026

Industrial SaaS for plastics & injection molding · converted from part-time to full-time Aachen, Germany (Remote)

- Built **Claire**, an injection-molding AI agent (Python, FastAPI, SQLAlchemy, Anthropic Claude) on an agent harness with tool calling and tenant-scoped data access; explains jobs, anomalies, downtime and experiment results, and flags data gaps instead of guessing. Live for all customers; showcased at Fakuma 2026.
- Built a voice shift-handover pipeline: speech-to-text plus LLM extraction turns spoken handovers into structured, searchable shift data.
- Led end-to-end development of 3 core SaaS projects as project owner (Svelte, NestJS); contributed PostgreSQL partitioning and vacuum tuning for 240M+ monthly time-series records in a multi-tenant architecture.
- Transformed a single-asset log into a unified contextual logbook linking 5+ entity types (assets, jobs, users, production cells) per entry, on a purpose-built PostgreSQL schema with real-time updates via Kafka.
- Directed a zero-downtime migration from plain-text to JSONB-rich content with Drizzle ORM; backfill scripts preserved 100% data integrity across all tenants.

Software Engineer (Staff Augmentation), *mobileLIVE (now Mlarteka)* 05/2024 – 10/2025

MonieCorp (Fintech Product) Canada (Remote)

- Developed modular UI components in React + TypeScript + shadcn, improving page load speed by 25% and reducing UI bugs by 40% after refactoring.
- Built 15+ SOLID-compliant admin-panel components, cutting client feedback cycles from 3 days to 24 hours.
- Built a multi-step form for tiered currency margins with validation; setup time cut from 20+ minutes to under 4 minutes per currency pair.
- Implemented CSP Level 3 with nonce-based whitelisting, blocking 100% of inline script execution attempts during penetration testing.

Staples Canada (mobileLIVE Client)

- Implemented Mixpanel tracking for 100,000+ users (product views, add-to-cart, checkout), helping the business team lift conversions by 12%.
- Built print-product customization end to end with the Canva API for the Staples Online Print Shop, increasing user engagement by 20%.
- Fixed 30+ production bugs within 4-hour SLAs, maintaining 99.9% uptime during peak sales.

Associate Software Engineer, *Code Huddle* 04/2022 – 05/2024

Associate Software Engineer 08/2022 – 05/2024

Product Development Intern 04/2022 – 06/2022

GYMYG: Virtual Gym Partner (US-Based Startup) Islamabad, Pakistan

- Built NestJS REST APIs with JWT RBAC, handling 10K DAU at under 200 ms P95 latency.
- Integrated Stripe monthly subscriptions, processing $15k+ in secure payments.
- Reduced video load times by ~30% with FFmpeg compression and AWS CloudFront edge delivery.
- Automated CI/CD with GitHub Actions and AWS CodePipeline, cutting deployment time by 40%.

OrderMe: QR Ordering Web App (Italian Startup MVP)

- Implemented a Firebase Auth + Firestore backend supporting 100+ daily orders.
- Built a shadcn/ui + Tailwind admin panel for menu creation, reducing owners' setup time by 60%.

Notable Freelance Projects

Pendfunds.com (MVP), *Contract Management & Mediation Software* 01/2024 – 09/2024

- Developed a full escrow payment flow with Stripe Connect and Optimus Bank merchant payouts in NestJS, onboarding 100+ users and handling hundreds of transactions.
- Automated deployments across development, staging and production with AWS Elastic Beanstalk and AWS CI/CD pipelines.
- Implemented payload encryption/decryption using PGP with RSA, keeping sensitive transaction data protected.

Education

Bachelor of Science in Software Engineering, *COMSATS University* 2018 – 2022

Islamabad, Pakistan

Skills

AI & LLMs

Python | FastAPI | SQLAlchemy | LLM Agents & Agent Harnesses | Tool Calling | RAG | LLM Extraction | Speech-to-Text | Anthropic Claude

Databases

PostgreSQL (partitioning, JSONB, tuning) | Drizzle ORM | TypeORM | Prisma | MySQL | MongoDB | Data Modelling

Cloud & DevOps

AWS (ECS, EC2, RDS, S3, CloudFront, CodePipeline) | Azure | Hetzner | Docker | GitHub Actions | CI/CD

Backend

TypeScript | Node.js | NestJS | Express.js | REST APIs | Kafka | Redis | Microservices | Stripe

Frontend

SvelteKit 5 | React.js | Next.js | Tailwind CSS | shadcn/ui | Redux Toolkit